

Dr Adam D Hines

Research Fellow · QUT Centre for Robotics

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Research profile

Robotician specialising in event-based vision, robotic localisation, and open-source research software. Research Fellow at the QUT Centre for Robotics, working on vision-based positioning for Roo-ver, Australia’s first lunar rover. Developer of EventCV, with research spanning embedded neuromorphic localisation, event-based place recognition, and reproducible evaluation workflows.

Professional experience

2026–present	Research Fellow QUT Centre for Robotics, Queensland University of Technology Working with Associate Professor Thierry Peynot on vision-based positioning for Roo-ver, Australia’s first lunar rover, as part of the QUT team.
2024–2026	Postdoctoral Research Fellow School of Natural Sciences, Macquarie University Computational modelling and biorobotics with Professor Andrew Barron; concurrent Visiting Fellow at QUT.
2023–2024	Research Fellow QUT Centre for Robotics, Queensland University of Technology Neuromorphic localisation with Professor Michael Milford and Dr Tobias Fischer.
2022	Visiting Associate Researcher / Fulbright Scholar University of Pennsylvania Research with Professor Roderic Eckenhoff.
2017–2019	Research Assistant Queensland Brain Institute, University of Queensland

Education

2019–2023	PhD in Neuroscience University of Queensland Fulbright Future and Queensland Scholar.
2013–2016	Bachelor of Medical Science (First Class Honours) University of New South Wales

Open-source software

EventCV

Developer of an open-source event-based computer vision library with a Python interface and Rust core. Work spans recording formats and live cameras, event representations, feature detection, motion estimation, simulation, reconstruction, visualisation, documentation, and developer tooling.

eventcv.net · [Documentation](#) · [Source code](#)

Event-LAB

Co-author of a framework for standardised evaluation of neuromorphic localisation methods, accepted to ICRA 2026.

[Paper and evaluation framework](#)

Teaching & supervision

- Primary supervisor for final-year engineering capstone students in QUT's EGH4001/2 courses, including project direction, mentorship, and assessment.
- Co-supervision of visiting PhD students, including work published at the IEEE Neuro Inspired Computational Elements conference.
- Designed and delivered a programming workshop for Master of Neuroscience students at the University of Queensland; the workshop was renewed for a second year.
- Completed Harvard Business School Online's Leadership Principles course in 2024.

Research funding

- 2026 — \$10,000 QUT Centre for Robotics early-career researcher grant for an event-based data capture system, commencing June 2026.
- 2022 — Fulbright Future and Queensland Scholarship, with competitive funding totalling \$25,000.

Honours & awards

- 2025 — Nominated for the QUT Vice-Chancellor's Award for Academic Excellence.
- 2022 — Fulbright Future and Queensland Scholarship.
- 2022 — Associate Faculty Member, H1 Connect.
- 2019–2023 — Young Science Ambassador outreach with Wonder of Science, recognised with a distinguished service award.
- 2019 — UQ Graduate School Scholarship.
- 2016–2021 — Student presentation prizes, including three-minute thesis and conference presentation awards.

Leadership, outreach & service

- 2026 — Associate Editor, IEEE/RSJ IROS Conference Paper Review Board.
- 2024–present — Steering committee member and co-chair roles, EMCR Brain Sciences Alliance.
- 2024–2025 — Flying Scientist, Office of the Chief Scientist of Queensland.
- 2024 — Organiser, Debates in Robotics Workshop at IEEE ICRA.
- 2020–2021 — Deputy chair and Queensland representative, Australasian Neuroscience Society Student Body Committee.
- Peer review for Science Robotics, IEEE Transactions on Robotics, ICRA, ICCV, ECCV, IEEE Robotics and Automation Letters, PLoS ONE, and Anesthesiology.

Presentations & public communication

- Invited seminars and presentations at MIT, Harvard, ETH Zurich, the University of Pennsylvania, UNSW, Technical University of Munich, and TU Delft.
- Research communication through the ABC, 2GB Radio, Australian Associated Press, and The Conversation.
- STEM outreach with schools in regional and remote Queensland through Flying Scientist and Wonder of Science.

Research collaborations

- Research collaborations across QUT, Macquarie University, TU Delft, and Western Sydney University, including bilateral work through a Templeton World Charity Foundation Research Hub.
- International research collaboration established during the Fulbright placement at the University of Pennsylvania.

Publications

The EventCV Library for Event-Based Robotic Vision

Adam D. Hines, Michael Milford, Tobias Fischer

2026 · arXiv · Preprint

arxiv.org/abs/2609.21330

Multi-viewpoint Geo-localization with Event Cameras

Adam D. Hines, Michael Milford, Tobias Fischer

2026 · arXiv · Preprint

arxiv.org/abs/2609.21219

EventGeM: Global-to-Local Feature Matching for Event-Based Visual Place Recognition

Adam D. Hines, Gokul B. Nair, Nicolás Marticorena, Michael Milford, Tobias Fischer

2026 · arXiv · Preprint

doi.org/10.48550/arXiv.2603.05807

Event-LAB: Towards Standardized Evaluation of Neuromorphic Localization Methods

Adam D. Hines, Alejandro Fontan, Michael Milford, Tobias Fischer

2026 · IEEE International Conference on Robotics and Automation (ICRA) · Accepted

doi.org/10.48550/arXiv.2509.14516

A neuromorphic model of the insect visual system for natural image processing

Adam D. Hines, Karin Nordström, Andrew B. Barron

2026 · arXiv · Preprint

doi.org/10.48550/arXiv.2602.06405

Learning complex stimuli with a honey bee brain: contrasting consequences on performance and flight behaviour from degrading configural versus featural information

M. G. Guiraud, **Adam D. Hines**, M. Jin, Andrew B. Barron

2026 · Research Square · Preprint

doi.org/10.21203/rs.3.rs-9487302/v1

A compact neuromorphic system for ultra-energy-efficient, on-device robot localization

Adam D. Hines*, Michael Milford, Tobias Fischer

2025 · Science Robotics · Published

doi.org/10.1126/scirobotics.ads3968

Publications · continued

Threshold Adaptation in Spiking Networks Enables Shortest Path Finding and Place Disambiguation

Robin Dietrich*, Tobias Fischer, Nicolai Waniek, Nico Reeb, Michael Milford, Alois Knoll, **Adam D. Hines**

2025 · IEEE Neuro Inspired Computational Elements (NICE) · Published

doi.org/10.1109/NICE65350.2025.11065806

Pixi: Unified Software Development and Distribution for Robotics and AI

Tobias Fischer, Wolf Vollprecht, Bas Zalmstra, Ruben Arts, Tim de Jager, Alejandro Fontan, **Adam D. Hines**, Michael Milford, Silvio Traversaro, Daniel Claes, Scarlett Raine

2025 · arXiv · Preprint

doi.org/10.48550/arXiv.2511.04827

Synapse-specific trapping of SNARE machinery proteins in the anesthetized Drosophila brain

Adam D. Hines, Amber B. Kewin, Matthew N. Van De Poll, Victor Anggono, Adekunle T. Bademosi, Bruno van Swinderen*

2024 · The Journal of Neuroscience · Published

doi.org/10.1523/JNEUROSCI.0588-23.2024

VPRTempo: A Fast Temporally Encoded Spiking Neural Network for Visual Place Recognition

Adam D. Hines, Peter G. Stratton, Michael Milford, Tobias Fischer

2024 · IEEE International Conference on Robotics and Automation (ICRA) · Published

doi.org/10.1109/ICRA57147.2024.10610918

Tracking Single Molecule Dynamics in the Adult Drosophila Brain

Adam D. Hines, Bruno van Swinderen*

2021 · eNeuro · Published

doi.org/10.1523/ENEURO.0057-21.2021

Pharmacological activation of IKr in models of long QT Type 2 risks overcorrection of repolarization

Matthew D Perry*, Chai-Ann Ng, Melissa M Mangala, Timothy Y M Ng, **Adam D Hines**, Whitney Liang, Michelle J O Xu, Adam P Hill, Jamie I Vandenberg

2020 · Cardiovascular Research · Published

doi.org/10.1093/cvr/cvz247

Turning to Drosophila for help in resolving general anesthesia

Bruno van Swinderen*, **Adam D Hines**

2020 · Proceedings of the National Academy of Sciences · Published

doi.org/10.1073/pnas.2012131117